02. Dataset Understanding

Know the schema, target, missingness and the limitations of the supplied data.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Dataset snapshot

Item	Observed
Rows / columns	1,000 × 20
Non-null rows per column	950; exactly 50 missing values in each column
Target	Loan_Approved: Yes / No
Numeric features	Applicant_Income, Coapplicant_Income, Age, Dependents, Credit_Score, Existing_Loans, DTI_Ratio, Savings, Collateral_Value, Loan_Amount, Loan_Term
Categorical features	Employment_Status, Marital_Status, Loan_Purpose, Property_Area, Education_Level, Gender, Employer_Category
Identifier	Applicant_ID

Feature meaning

- Income and co-applicant income: repayment capacity indicators; validate currency and time basis.
- Credit_Score: creditworthiness proxy; the source does not document score scale or bureau.
- DTI_Ratio: debt-to-income ratio; higher values often imply greater repayment burden, but the definition is not supplied.
- Existing_Loans, Savings, Collateral_Value, Loan_Amount and Loan_Term: indebtedness, reserves, security, requested principal and repayment horizon.
- Employment_Status and Employer_Category: employment context; check overlap and consistency.
- Loan_Purpose and Property_Area: application context and geography.
- Age, Gender, Marital_Status, Education_Level and Dependents: demographic or household attributes requiring careful legal/fairness review.

Data quality

- Every column has 50 nulls (5% of records), including the target. This is unusual and may indicate synthetic row-level missingness.
- Numeric ranges are bounded in the sample: Credit_Score 550–799, DTI_Ratio 0.10–0.60, Loan_Term 12–84 months.
- Applicant_ID is an identifier and should not be used as a predictive signal.
- The dataset appears small and synthetic/demo-like; there is no data dictionary, provenance, time period, currency or policy definition.

Critical target issue

The notebook imputes missing Loan_Approved values using the most frequent class, which is No. This manufactures target labels for 50 applicants rather than leaving them unknown. Those rows then enter model training/testing as if the labels were observed. A sound workflow should exclude rows with unknown target labels from supervised training and separately investigate why labels are missing.

Data questions before modelling

- Does the target represent final approval, pre-approval or a historical decision under a specific policy?
- Are features measured before the decision, or do any encode post-decision information?
- Why are exactly 50 values missing in every field? Is there a common set of 50 partially blank records?
- What do DTI, savings and collateral units mean?
- Are demographic features legally permissible and appropriate for the intended jurisdiction?